

PHYS-E0510 – Advanced Computational Methods in Physics D

Session 2: Exact Diagonalization of the Heisenberg Spin Chain

Companion notebook: `quantum_magnetism.ipynb` **Prerequisites:** Session 1 (second quantization and the superexchange derivation of the Heisenberg model)

Contents and goals

Last time we ended on a cautionary note: mean-field theory, built entirely out of product states, cannot represent the true ground state of the Hubbard dimer, an entangled singlet that spontaneously breaks nothing at all. Today we stop approximating. Instead of guessing a variational ansatz, we are going to write down the many-body Heisenberg Hamiltonian that emerges from superexchange, represent it as an honest matrix acting on the full 2^L -dimensional many-body Hilbert space, and diagonalize it exactly. Along the way we will learn to read a many-body spectrum: how the many-body gap tells us whether a spin chain is gapped or gapless, how a field or a coupling constant can be dialed to drive level crossings and even phase transitions, how symmetries let us block-diagonalize the Hamiltonian so that “exact” does not immediately mean “impossible,” and how correlators, both static and dynamical, let us extract order and excitations directly from the wave function. We will close by asking whether the excitations of a quantum magnet admit their own single-particle description, in terms of emergent fermionic quasiparticles called spinons. Everything here is numerically exact, no product-state restriction anywhere, and that honesty comes at the price we will keep bumping into all session: the Hilbert space grows as 2^L , which is exactly the wall that motivates the tensor-network methods we build starting in Session 4.

Learning outcomes

By the end of this session, you should be able to:

- Build the Heisenberg Hamiltonian as an explicit matrix on the many-body product-state basis and diagonalize it exactly for a finite spin chain.
- Use the finite-size scaling of the many-body gap, power-law versus saturating, to tell a gapless spin chain from a gapped one.
- Exploit conserved quantum numbers, total spin and S_{tot}^z , to block-diagonalize the Hamiltonian and explain level crossings and magnetization curves in a field.
- Compute static and dynamical spin correlators and connect them to an auxiliary-fermion (spinon) mean-field description of the excitations.

1 From the Hubbard dimer to a genuine quantum magnet

Let us start by reminding ourselves what a symmetry-broken classical magnet actually looks like at the microscopic level, since it is the natural foil for what comes next. Take the simplest possible magnetic Hamiltonian, an Ising coupling between two spins,

$$\mathcal{H}_{\text{Ising}} = J S_1^z S_2^z, \quad J > 0. \quad (1)$$

There is no S^x or S^y anywhere, so you can just as well think of $S_i^z = \pm \frac{1}{2}$ as a classical Ising variable. Minimizing the energy is trivial: we want S_1^z and S_2^z to have opposite sign, and there are exactly two ways to do that, $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. Both are legitimate ground states, both break time-reversal symmetry (flipping every spin sends one into the other), and each one individually has a finite, sharply defined local moment $\langle S_i^z \rangle = \pm \frac{1}{2}$. This is a classical magnet in miniature: a genuinely symmetry-broken state with a finite magnetization at every site.

Now let us restore the full quantum mechanics of the spin operators and couple all three components,

$$\mathcal{H}_{\text{Heis}} = J \mathbf{S}_1 \cdot \mathbf{S}_2 = J (S_1^x S_2^x + S_1^y S_2^y + S_1^z S_2^z). \quad (2)$$

This is exactly the superexchange Hamiltonian we derived last session from the strong-coupling limit of the Hubbard dimer, with $J = 4t^2/U > 0$. We already know, from Session 1, that its exact ground state is the singlet $|\Psi_{\text{singlet}}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$: a single, non-degenerate state, with $\langle S_i^z \rangle = 0$ on every site, that cannot be written as any product state. This is what we will from now on call a genuine **quantum magnet**: unlike the Ising ground states, this state does not break time-reversal symmetry at all, and it is entangled in an essential way. Our job today is simply to compute this kind of ground state honestly, for chains longer than two sites, and to understand everything we can read off from it and from the excited states that sit above it.

2 Exact diagonalization: basis, matrix, and a worked dimer

Generalize Eq. (2) to a chain of L spin- $\frac{1}{2}$'s with nearest-neighbor exchange,

$$\mathcal{H}_{\text{Heis}} = J \sum_{i=1}^{L-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \quad J > 0 \text{ (antiferromagnetic)}. \quad (3)$$

Regardless of how complicated J_{ij} ends up being on some lattice, solving this Hamiltonian exactly always reduces to the same three-step recipe: pick a basis of the many-body Hilbert space, write $\mathcal{H}$ as a matrix in that basis, and diagonalize it. The natural basis is the set of all product states $|\sigma_1 \sigma_2 \cdots \sigma_L\rangle$ with $\sigma_i \in \{\uparrow, \downarrow\}$: two states for $L = 1$, four for $L = 2$, eight for $L = 3$, and in general 2^L states for L sites. This is precisely the exponential growth of the many-body Hilbert space we have already met once, in Session 1's discussion of the Hubbard model; here it shows up again because a spin- $\frac{1}{2}$ chain is nothing but a projected, single-occupied corner of that same fermionic Hilbert space.

To build the matrix we need the matrix elements of $\mathbf{S}_i \cdot \mathbf{S}_j$, and it is most convenient to split it into a diagonal and an off-diagonal piece using the raising and lowering operators $S^\pm = S^x \pm iS^y$,

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+). \quad (4)$$

The first term is diagonal in our product basis: it just multiplies each basis state by $\pm \frac{1}{4}$ depending on whether σ_i, σ_j are parallel or antiparallel. The second term is the one that actually moves us around the basis: $S_i^+ S_j^-$ flips an antiparallel pair $\downarrow_i \uparrow_j \rightarrow \uparrow_i \downarrow_j$ and annihilates everything else, so it only ever connects basis states that differ by swapping one $\uparrow\downarrow$ pair on a bonded pair of sites. This is why the Heisenberg matrix, even though it lives in a 2^L -dimensional space, is *sparse*: each row has at most $L - 1$ off-diagonal entries, one per bond, which is exactly the kind of structure that iterative solvers like Lanczos exploit to reach much bigger L than a naive dense diagonalization ever could.

Worked example: the dimer, twice over. Let us see this explicitly for $L = 2$. In the basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$, the stretched states $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ have no partner to flip into, so $\mathcal{H}_{\text{Heis}}|\uparrow\uparrow\rangle = \frac{J}{4}|\uparrow\uparrow\rangle$ and likewise for $|\downarrow\downarrow\rangle$. On the remaining two-dimensional subspace $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$, Eq. (4) gives the 2×2 block

$$\mathcal{H}_{\text{Heis}}|_{\{\uparrow\downarrow, \downarrow\uparrow\}} = J \begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix} \implies E_{\pm} = J \left(-\frac{1}{4} \pm \frac{1}{2} \right), \quad |\pm\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle \pm |\downarrow\uparrow\rangle). \quad (5)$$

So $E_- = -\frac{3}{4}J$ for the antisymmetric combination, exactly the singlet we already knew, and $E_+ = \frac{1}{4}J$ for the symmetric combination, degenerate with $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ at the same energy $\frac{J}{4}$: together these three states form the **triplet**. Diagonalizing a 4×4 matrix by hand already reproduces the full singlet/triplet structure.

There is a second, quicker way to get the same answer, and it is worth knowing because it generalizes. Define the total spin $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$, so that $\mathbf{S}_{\text{tot}}^2 = \mathbf{S}_1^2 + \mathbf{S}_2^2 + 2\mathbf{S}_1 \cdot \mathbf{S}_2$, and hence

$$\mathcal{H}_{\text{Heis}} = J \mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{J}{2} [S_{\text{tot}}(S_{\text{tot}} + 1) - \frac{3}{4} - \frac{3}{4}], \quad (6)$$

using $\mathbf{S}_i^2 = \frac{3}{4}$ for spin- $\frac{1}{2}$. Since $\mathbf{S}_{\text{tot}}^2$ has eigenvalues $S_{\text{tot}}(S_{\text{tot}} + 1)$ with $S_{\text{tot}} = 0$ or 1 for two spin- $\frac{1}{2}$'s, Eq. (6) immediately gives $E(S_{\text{tot}} = 0) = -\frac{3}{4}J$ and $E(S_{\text{tot}} = 1) = \frac{1}{4}J$, matching Eq. (5) exactly. Notice, too, why the notebook's ground-state energy happens to equal the sum of the separately computed Ising and XY ground energies: on the two-dimensional subspace $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$, the Ising piece $S_1^z S_2^z$ is literally proportional to the identity, $-\frac{1}{4}\mathbb{I}$, since *both* basis states carry the same $S_1^z S_2^z$ eigenvalue. Minimizing the sum of a constant and the XY block is the same as minimizing each separately and adding, which is why the two energies happen to add up here. This is special to $L = 2$: for longer chains the Ising term is no longer a multiple of the identity on the subspace the XY term mixes, and the two pieces generically no longer minimize on the same state.

✕ Computational companion

Open `quantum_magnetism.ipynb` and work through the “Basics” block: “Creating a spin_chain object,” “Creating a Hamiltonian,” “Computing the ground state energy,” and “Computing expectation values.” Build the $L = 2$ Heisenberg Hamiltonian directly from the spin operators, diagonalize it, and check that you recover $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = \frac{1}{4}J$ numerically. Then replace S^x by S^y in the expectation-value cell and think about why nothing changes.

3 The many-body spectrum and the many-body gap

Now let us stop looking at $L = 2$ by hand and let the computer diagonalize Eq. (3) for increasing L . The number of eigenvalues you get out is, of course, exactly the dimension of the matrix, 2^L : two for $L = 2$, sixteen for $L = 4$, sixty-four for $L = 6$, and so on, growing exponentially with system size, exactly like the basis itself. In practice, though, we almost never care about the whole spectrum. Any real experiment at low temperature only probes the states closest to the ground state, so from now on we will systematically focus on the bottom of the many-body spectrum: the ground state E_0 and the handful of low-lying excited states just above it.

The single most useful number we can read off from this part of the spectrum is the **many-body**

gap,

$$\Delta(L) \equiv E_1(L) - E_0(L), \quad (7)$$

the energy cost of the cheapest excitation above the ground state. For the uniform Heisenberg chain, $\Delta(L)$ shrinks steadily as L grows, and it does so following a power law rather than saturating to a finite value: this is the hallmark of a **gapless** system, one whose excitations become arbitrarily cheap in the thermodynamic limit. Where does this power law come from? Qualitatively, from exactly the same free-fermion logic we will use in Section 10: replace each spin operator by an auxiliary (Abrikosov) fermion $f_{i\sigma}$ and mean-field decouple the exchange term, and $\mathcal{H}_{\text{Heis}}$ turns into a free-fermion **spinon** Hamiltonian whose band is gapless, with Fermi points at $k_F = \pm\pi/2$. The cheapest excitations are then particle-hole pairs created right at those Fermi points, which disperse *linearly* in momentum, and a linearly dispersing excitation on a ring of L sites is exactly a one-dimensional particle-in-a-box problem: its levels are spaced by

$$\Delta(L) \sim \frac{\pi v_s}{L}, \quad (8)$$

with v_s the velocity of the low-energy excitations, i.e. a genuine $1/L$ power law rather than a finite gap. (Numerically, the finite-size data is well fit by $\Delta(L) \approx \pi^2 J/(2L)$; pinning down v_s this precisely is a job for the exact Bethe-ansatz solution, not for mean-field theory, which gets the qualitative power law right but not the precise coefficient.) Keep this “particle in a box” picture in mind: we will meet it again, in a much more literal form, when we look at dynamical correlators in Section 9, and again when we build the explicit spinon band structure behind this argument in Section 10.

✕ Computational companion

Section “Many-body energies of the Heisenberg model.” Diagonalize the uniform chain for $L = 2, 4, 6, 8$ and plot the full spectrum each time: check that the number of eigenvalues really is 2^L . Then plot only $E_1(L) - E_0(L)$ and see the gap shrink with L ; think about what the biggest L is that you could still store the full spectrum for on your own machine.

4 The XXZ model: from classical order to quantum entanglement

Section 1 presented the Ising model and the Heisenberg model as two separate Hamiltonians, but it is much more illuminating to see them as the two endpoints of a single family. Take

$$\mathcal{H}_{\text{XXZ}} = \sum_i \left[J_{xy} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y) + J_z S_i^z S_{i+1}^z \right], \quad J_z \equiv 1 \text{ fixed}, \quad (9)$$

and dial J_{xy} from 0 to 1. At $J_{xy} = 0$ we are exactly back at the classical Ising model of Eq. (1), generalized to L sites: in the thermodynamic limit its ground state is doubly degenerate, the Néel-ordered $|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ and its time-reversed partner, each carrying a finite, symmetry-broken staggered moment. At $J_{xy} = 1$ we recover the isotropic Heisenberg model of Eq. (3), whose ground state, as we already know from the dimer, is unique and carries no local moment at all: a purely quantum magnet. So J_{xy} literally interpolates between a classical, symmetry-broken magnet and a quantum, entangled one, and the natural question is: how does the ground state actually get from one description to the other as we sweep J_{xy} ?

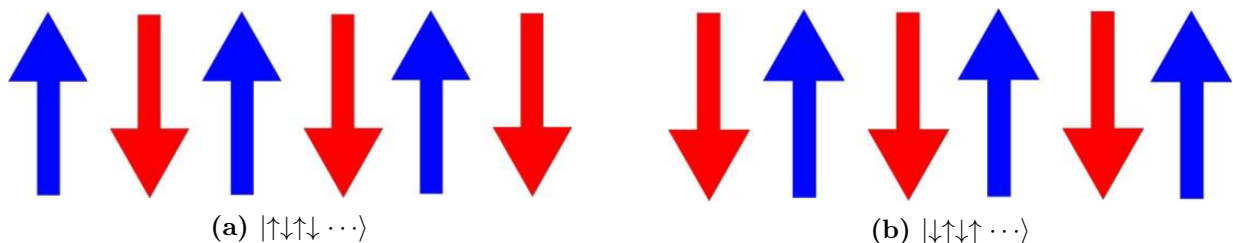


Figure 1: The two classical ground states of Eq. (9) at the Ising endpoint $J_{xy} = 0$. Each is a product state with a definite S^z on every site, each carries a finite staggered moment, and the two are exchanged by time reversal. In the thermodynamic limit they are exactly degenerate and the system picks one of them, which is what symmetry breaking means here. On a finite chain they are instead split by a tiny amount, and watching that splitting grow as $J_{xy} \rightarrow 1$ is how the crossover to the unique, entangled Heisenberg ground state shows up in the spectrum.

The finite-size spectrum answers this beautifully. At small J_{xy} , the two classical Néel states are not exactly degenerate on a finite chain (only a genuinely infinite chain can spontaneously break a symmetry), but they sit extremely close together, split by a tiny amount that we can literally watch on the screen: this is the finite-size ghost of the classical double degeneracy. As we increase J_{xy} toward 1, this splitting grows continuously, and for any finite L that growth eventually turns into a finite, clearly non-degenerate gap once J_{xy} gets close enough to 1. What changes as we make L bigger is exactly how close to $J_{xy} = 1$ this crossover sits: for larger and larger chains, the two low-lying states stay nearly degenerate over a longer and longer stretch of J_{xy} , and the crossover sharpens toward the exact, thermodynamic-limit transition of the XXZ chain, which the Bethe-ansatz solution places precisely at the isotropic, $SU(2)$ -symmetric point $J_{xy} = J_z$. In other words: just by watching how two eigenvalues in a finite-size spectrum move and split as we tune a coupling constant, we can already diagnose a genuine quantum phase transition, well before we ever reach the thermodynamic limit itself. This is a technique we will lean on again and again in the rest of the course.

✕ Computational companion

Section “From classical to quantum magnetism.” Plot the low-lying spectrum of $\mathcal{H}_{\text{XXZ}}$ as a function of J_{xy} at fixed L : identify the near-degenerate pair at small J_{xy} and watch it split as $J_{xy} \rightarrow 1$. Then increase L and see how the splitting crossover sharpens and moves, tracking the exact thermodynamic-limit transition at the isotropic point.

5 Good quantum numbers: total spin, the Zeeman field, and block-diagonalization

Section 2 used the identity $\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2}[\mathbf{S}_{\text{tot}}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2]$ as a shortcut for the dimer, but it is worth asking why it worked: it worked because $\mathbf{S}_{\text{tot}}^2$ and S_{tot}^z commute with $\mathcal{H}_{\text{Heis}}$. This is not special to two sites. Because Eq. (3) is built entirely out of dot products $\mathbf{S}_i \cdot \mathbf{S}_j$, it is invariant under any global spin rotation, and Noether’s theorem in its operator form tells us immediately that the total spin $\mathbf{S}_{\text{tot}} = \sum_i \mathbf{S}_i$ is conserved for *any* L and any choice of exchange couplings J_{ij} :

$$[\mathcal{H}_{\text{Heis}}, \mathbf{S}_{\text{tot}}^2] = 0, \quad [\mathcal{H}_{\text{Heis}}, S_{\text{tot}}^z] = 0. \quad (10)$$

So every eigenstate of $\mathcal{H}_{\text{Heis}}$ can be labeled by a definite total spin S and a definite projection m , exactly the singlet ($S = 0$) and triplet ($S = 1$) labels we used for the dimer, now promoted to good quantum numbers for a chain of any length.

The Zeeman field does not mix states. Let us now add a uniform magnetic field,

$$\mathcal{H}(B) = \mathcal{H}_{\text{Heis}} - B S_{\text{tot}}^z. \quad (11)$$

This term explicitly singles out the z -axis, so it breaks the full $\text{SU}(2)$ spin-rotation symmetry down to rotations about z alone. But look closely at what it actually does to our two conserved quantities. It trivially commutes with S_{tot}^z , since it is built out of nothing else. And because $[\mathbf{S}_{\text{tot}}^2, S_{\text{tot}}^z] = 0$ is a completely general identity of angular momentum algebra, $\mathcal{H}(B)$ *also* commutes with $\mathbf{S}_{\text{tot}}^2$. So both quantum numbers survive even in the presence of the field: adding B cannot mix any eigenstates of $\mathcal{H}_{\text{Heis}}$ into each other, it can only shift each one's energy by $-Bm$. This is exactly why the field-dependent spectrum you see in the notebook is a set of perfectly straight lines rather than curves: each state with a definite m shifts linearly, $E(B) = E(0) - Bm$, and states only bend or avoid one another once you add a term, like a transverse field, that fails to commute with S_{tot}^z .

Applied to the dimer: the singlet has $m = 0$ and stays completely flat, $E_{\text{singlet}}(B) = -\frac{3}{4}J$. The triplet's three members split according to their own $m = -1, 0, +1$: the $m = 0$ member is also completely flat, at $E = \frac{1}{4}J$, while the $m = \pm 1$ members disperse as $E = \frac{1}{4}J \mp B$. Those are precisely the two “flat lines” you can spot in the notebook's zero-field spectrum: they sit at different energies, separated by exactly the many-body gap $\Delta = J$ of Eq. (7), and what distinguishes them is not m , which is 0 for both, but the total spin S , 0 for one and 1 for the other.

Block-diagonalizing a long chain. The real payoff of Eq. (10) is computational. Since $S_{\text{tot}}^z = \sum_i S_i^z$ is conserved, the full $2^L \times 2^L$ Hamiltonian never actually mixes basis states with different numbers of up spins: if $n_{\uparrow}$ counts the up spins, then $m = n_{\uparrow} - L/2$ is fixed within each block, and $\mathcal{H}_{\text{Heis}}$ is block-diagonal with blocks of dimension

$$\dim(m) = \binom{L}{L/2 + m}. \quad (12)$$

The largest block sits at $m = 0$ (for even L), with $\dim(0) = \binom{L}{L/2} \sim 2^L / \sqrt{\pi L/2}$ by Stirling's approximation: still exponential in L , but a factor $\sim \sqrt{L}$ smaller than the full Hilbert space, and, crucially, diagonalizable completely independently of every other block. If the chain is also periodic, translational invariance gives us a second conserved quantity, the total momentum k , and combining S_z -sectors with momentum sectors shrinks the largest block by roughly another factor of L . This combination, exploiting every available conserved quantity together with the sparse-matrix Lanczos techniques of Section 2, is exactly how a code like `dmrgpy` manages to push exact diagonalization out to the $L \lesssim 20$ spins we will be working with all session; without it, $2^{20} \approx 10^6$ would already be a serious dense-matrix computation, and $L = 30$ would simply be out of reach.

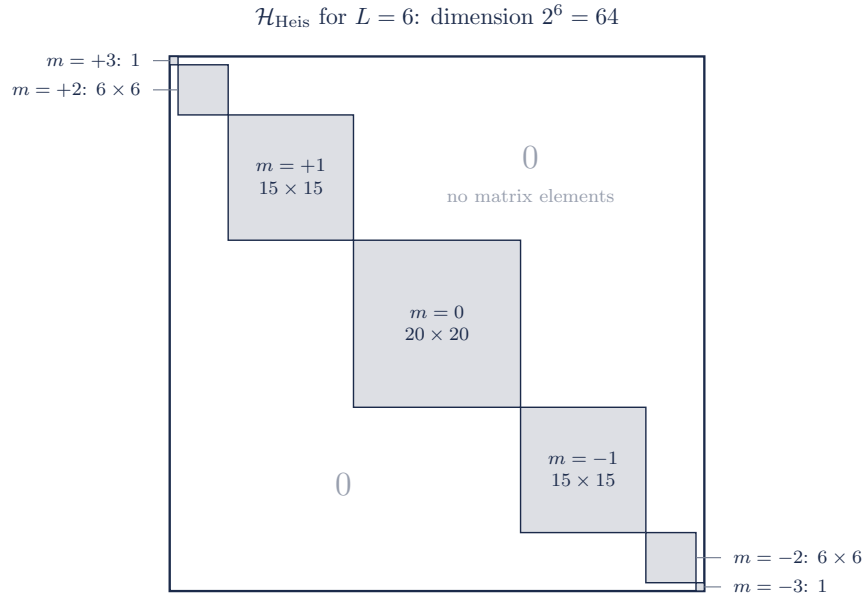


Figure 2: Block structure of $\mathcal{H}_{\text{Heis}}$ in the S_{tot}^z basis, drawn for $L = 6$ with the blocks to scale. Because S_{tot}^z is conserved, the Hamiltonian never connects basis states with different numbers of up spins, so the $2^L \times 2^L$ matrix falls apart into one block per magnetization sector, of dimension $\binom{L}{L/2+m}$ as in Eq. (12), with nothing at all outside them. Each block is diagonalized completely independently of the others, and even the largest one, at $m = 0$, is already smaller than the full space by a factor $\sim \sqrt{L}$.

✕ Computational companion

Sections “Energies of a Heisenberg model as a function of the external magnetic field” and “Quantum numbers in the Heisenberg model.” Plot the dimer spectrum against B and check the linear, non-mixing evolution predicted by Eq. (11); identify which line is the singlet and which is the $m = 0$ triplet member, and confirm that they are exactly $\Delta = J$ apart at $B = 0$. Then think about what field-dependent term would bend the lines instead of keeping them straight.

6 Level crossings and magnetization curves

Equation (11) does more than split levels: because different m -sectors disperse with different slopes, they can cross, and a crossing at the very bottom of the spectrum is a genuine **level crossing of the ground state**. For the dimer this is easy to pin down exactly: the ground state is the singlet, at $-\frac{3}{4}J$, until the $m = +1$ triplet branch $E = \frac{1}{4}J - B$ drops below it, which happens once

$$\frac{1}{4}J - B = -\frac{3}{4}J \implies B = J. \quad (13)$$

Below $B = J$ the ground state is the singlet, with zero magnetization; above $B = J$ it is the fully polarized triplet member $|\uparrow\uparrow\rangle$, with saturated magnetization $\langle S_{\text{tot}}^z \rangle = 1$. This is the simplest possible example of a field-driven quantum phase transition: a single, sharp level crossing between a sector with $S = 0$ and a sector with $S = 1$.

Longer chains show the same phenomenon, just repeated many times over. As B grows from

zero, the ground state hops successively through sectors of increasing total spin, $S = 0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$, each hop showing up as an abrupt jump in the ground-state magnetization $\langle S_{\text{tot}}^z \rangle$: a magnetization staircase. Because bigger chains pack their low-lying S -sectors closer together in energy, these crossings happen at progressively smaller fields as L grows, and in the thermodynamic limit the staircase smooths out into the continuous magnetization curve of an infinite antiferromagnetic chain. Eventually, once B exceeds roughly the bandwidth of the low-energy spectrum ($B \gtrsim zJ$, with z the coordination number; for a chain, $z = 2$ and the saturation field is exactly $B_{\text{sat}} = 2J$), the ground state saturates completely: every spin points along the field, and this fully polarized state is in fact an *exact* eigenstate of $\mathcal{H}_{\text{Heis}}$ itself, a rare case where the many-body ground state really is a simple product state, since there is nothing left for the exchange term to do once every spin is already maximally aligned. Sweeping $B \rightarrow -B$ reproduces the identical staircase with the opposite sign of magnetization, exactly the time-reversal partner of the $B > 0$ curve.

It is also worth looking directly at the excited states in a small field, before any level crossing has happened. The ground state stays a total- $S = 0$ singlet with zero magnetization on every site. The first excited states, the $S = 1$ triplet members, do carry a finite total magnetization, but that magnetization is not localized anywhere: it is smeared out over the whole chain, one more signature that these are genuinely delocalized many-body excitations rather than a single flipped spin sitting at one site. Higher excited states repeat this pattern with different sign structures, several of them related to each other by time reversal, and this whole family of spread-out, roughly particle-in-a-box-like excitations is exactly what the dynamical correlator of Section 9 will let us resolve in detail.

✕ Computational companion

Sections “Level crossings in the Heisenberg model,” “Many-body wavefunctions in a magnetic field,” “Level crossings as a function of the magnetic field,” and “Magnetization of moderately large Heisenberg model.” Confirm the dimer’s crossing at $B = J$ from Eq. (13), then move to $L = 6$ and $L = 10$ chains and track the full magnetization staircase, its saturation field, and its mirror symmetry under $B \rightarrow -B$.

7 Gapped versus gapless: the dimerized chain

Section 3 told us the uniform Heisenberg chain is gapless, $\Delta(L) \rightarrow 0$ as a power law. Let us now build a chain that is not, by alternating the exchange coupling bond by bond,

$$\mathcal{H}_{\text{dimer}} = \sum_i J_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \quad J_i = J [1 + (-1)^i \delta], \quad (14)$$

so odd and even bonds alternate between $J(1 + \delta)$ and $J(1 - \delta)$. At $\delta = 0$ we recover the uniform, gapless chain. At the opposite extreme $\delta \rightarrow 1$, the weak bonds vanish entirely and the chain falls apart into $L/2$ completely decoupled dimers, each one exactly the $L = 2$ problem of Section 2: the exact ground state is simply a product of singlets, one per strong bond, and the cheapest excitation is a single dimer flipping from its singlet into its triplet, costing exactly $\Delta_\infty = J(1 + \delta)$, the very same singlet-triplet gap we computed in Eq. (5). Turning the weak bond back on perturbs this picture, letting the singlets talk to each other and giving the triplet excitation some dispersion, but it does not close the gap: for any finite δ , no matter how small,

the dimerized chain remains gapped, with a finite Δ_∞ that shrinks smoothly to zero only as $\delta \rightarrow 0$.

This qualitative difference shows up very cleanly in how $\Delta(L)$ approaches its thermodynamic-limit value. For the gapless uniform chain, we already saw the slow, power-law approach of Eq. (8): even $L \sim 20$ is not enough to convince yourself the gap is really heading to zero. For the dimerized chain, a finite gap sets its own length scale, the correlation length $\xi \sim v_s/\Delta_\infty$, and finite-size corrections to the gap are suppressed exponentially, $\Delta(L) - \Delta_\infty \sim e^{-L/\xi}$, rather than as a power law. That is why, numerically, a dimerized chain's gap already looks converged by $L \sim 12$ –14, while the uniform chain's gap is still visibly drifting at the same size: the presence or absence of a length scale ξ is precisely what separates fast, exponential finite-size convergence from slow, power-law convergence.

✕ Computational companion

Section “Gap of a many-body spin chain.” Compute $\Delta(L)$ for the uniform chain ($\delta = 0$) and for a dimerized chain ($\delta > 0$), and compare how quickly each one saturates as you increase L ; check that the dimerized gap has essentially converged well before the uniform one has.

8 Diagnosing the gap: impurities and non-local correlators

Comparing gaps at different L works, but it needs several system sizes side by side. Is there a way to diagnose gapped versus gapless behavior from a *single* calculation? Yes: look at how the system responds in space rather than how its energy responds in size.

A magnetic impurity. Pin a small local field on one site, $\mathcal{H} \rightarrow \mathcal{H}_{\text{Heis}} - h S_{i_0}^z$, and measure the induced magnetization profile $\langle S_j^z \rangle$ as a function of distance $r = |j - i_0|$ from the impurity. For the gapless uniform chain, this response decays *algebraically*, $\langle S_j^z \rangle \sim r^{-1}$, the standard hallmark of critical, scale-free correlations in a one-dimensional gapless system. For the gapped dimerized chain, the very same measurement decays *exponentially*, $\langle S_j^z \rangle \sim e^{-r/\xi}$, with the same correlation length ξ that controlled the finite-size convergence of the gap in Section 7. So the way a localized perturbation “heals” as you move away from it already tells you, from a single system size, whether the underlying chain is gapped or gapless.

Non-local static correlators. We can get exactly the same information without introducing any impurity at all, by computing the connected two-point correlator

$$C(i, j) = \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle. \quad (15)$$

In a genuine quantum magnet $\langle \mathbf{S}_i \rangle = 0$ everywhere, so this reduces to the bare correlator $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$; for a classical, product-state magnet, on the other hand, $C(i, j)$ vanishes identically, since a product state trivially factorizes the expectation value. So a nonzero $C(i, j)$ is itself a direct signature of entanglement beyond any mean-field description, echoing exactly the failure mode we uncovered for the Hubbard dimer in Session 1. Formally, this static correlator is tied to the impurity response of the previous paragraph by linear response (a Kubo-formula argument): the induced magnetization at j from a small field at i is, to leading order, proportional to $C(i, j)$

itself. It is therefore no surprise that $C(i, j)$ shows exactly the same dichotomy: power-law decay for the gapless chain, exponential decay, with a rate set by δ , for the dimerized one. The bigger the dimerization, the bigger the gap, and the faster $C(i, j)$ falls off, a clean, single-snapshot diagnostic for how gapped a quantum magnet really is.

✕ Computational companion

Sections “Magnetization of a Heisenberg model with a magnetic impurity,” “Magnetization of a dimerized Heisenberg model with a magnetic impurity,” and “Non-local static correlators.” Compare the decay of the induced magnetization for the uniform and dimerized chains, then compute Eq. (15) directly for several values of the dimerization and check that faster decay tracks a bigger gap.

9 Dynamical correlators, and a first look at spin-1

So far we have only used static, equal-time expectation values. A **dynamical spin correlator** carries strictly more information, because it resolves excitations by their energy as well as their location. Define

$$S(i, \omega) = \langle 0 | S_i^z \delta(\omega - (\mathcal{H} - E_0)) S_i^z | 0 \rangle. \quad (16)$$

This looks abstract until we insert a resolution of the identity in the exact eigenbasis of $\mathcal{H}$, $\mathbb{1} = \sum_n |n\rangle \langle n|$, which turns Eq. (16) into the Lehmann representation

$$S(i, \omega) = \sum_n |\langle n | S_i^z | 0 \rangle|^2 \delta(\omega - (E_n - E_0)). \quad (17)$$

Read this as a spectroscopy experiment: it says that if you inject a spin flip at site i with energy ω , you get a nonzero response exactly when there exists an eigenstate $|n\rangle$ at energy $E_n = E_0 + \omega$ that is actually connected to the ground state by a local spin flip, weighted by how big that overlap $\langle n | S_i^z | 0 \rangle$ is.

For the spin- $\frac{1}{2}$ uniform chain, $S(i, \omega)$ shows a sequence of discrete peaks, spread out over the whole chain, separated by gaps: exactly the particle-in-a-box quantization we anticipated in Eq. (8). As L grows, these peaks soften toward lower ω , tracking the shrinking many-body gap of Section 3, and more of them appear as the low-energy spectrum becomes denser.

A first look at spin-1. If we redo this same calculation for a spin-1 Heisenberg chain instead of spin- $\frac{1}{2}$, the profile of $S(i, \omega)$ looks qualitatively different. This is because the two cases sit in genuinely different phases: the spin- $\frac{1}{2}$ chain is gapless in the thermodynamic limit, as we have seen throughout this session, while the spin-1 chain is what is called a **topological quantum magnet**, with a full gap in the bulk but gapless, fractionalized spin- $\frac{1}{2}$ excitations pinned to the two ends of an open chain. The underlying reason half-integer and integer spin chains behave so differently is the Lieb–Schultz–Mattis theorem, which forbids a translationally invariant, SU(2)-symmetric chain with half-integer spin per site from having a unique, gapped ground state, while placing no such restriction on integer spin. In our small finite-size dynamical correlators we can already see hints of this: a low-energy excitation that lives mostly at the two edges and gets progressively softer as L grows, sitting well below a much larger, roughly size-independent bulk gap. Resolving this fractionalization cleanly, however, needs system sizes well beyond what exact diagonalization can reach, which is exactly the job tensor networks will do for us in Session 4.

✕ Computational companion

Section “Dynamical correlators.” Compute $S(i, \omega)$ for the spin- $\frac{1}{2}$ Heisenberg chain, make the system smaller and watch the discrete peaks move, and add a magnetic field to see how the Zeeman shifts of Section 5 show up directly in the correlator’s peak positions.

10 Quasiparticles for the ground state: Abrikosov fermions and spinon mean-field theory

Everything we have computed so far treats the ground state and its excitations as intrinsically many-body objects, with no single-particle description in sight. But we can ask the same question we asked in Session 1 about the Hubbard model: is there some auxiliary single-particle Hamiltonian whose spectrum reproduces, at least qualitatively, the low-energy excitations of $\mathcal{H}_{\text{Heis}}$? The answer is yes, and it comes from representing the quantum spin operators themselves in terms of auxiliary fermions.

The Abrikosov (auxiliary) fermion representation. Introduce a fermion $f_{i\sigma}$ per site per spin flavor and write

$$S_i^\alpha = \frac{1}{2} \sum_{\sigma\sigma'} f_{i\sigma}^\dagger \tau_{\sigma\sigma'}^\alpha f_{i\sigma'}, \quad \alpha = x, y, z, \quad (18)$$

with τ^α the Pauli matrices, subject to the local constraint $\sum_\sigma f_{i\sigma}^\dagger f_{i\sigma} = 1$ on every site: exactly one auxiliary fermion, no more, no fewer. This constraint is not optional bookkeeping, it is what makes the representation exact rather than approximate: an unconstrained spinful fermion per site has four states, empty, singly occupied ($\uparrow$ or $\downarrow$), and doubly occupied, but the constraint removes the empty and doubly occupied configurations and leaves exactly the two singly-occupied states $\{|\uparrow\rangle, |\downarrow\rangle\}$, which is precisely the spin- $\frac{1}{2}$ Hilbert space we started with. These auxiliary fermions are emphatically not electrons; they never hop in or out of the constrained subspace, they simply parameterize the same two-dimensional local Hilbert space in fermionic language.

Substituting Eq. (18) into $\mathbf{S}_i \cdot \mathbf{S}_j$ and using the constraint on each site, one finds the standard identity

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2} \chi_{ij}^\dagger \chi_{ij} + \frac{1}{4} n_i n_j, \quad \chi_{ij} \equiv \sum_\sigma f_{i\sigma}^\dagger f_{j\sigma}, \quad (19)$$

where χ_{ij} is the fermionic **bond operator**. This turns the quartic spin exchange into a quartic fermion interaction, exactly the kind of term that Session 1 taught us how to handle: decouple it at the mean-field level. Introduce the self-consistent bond amplitude $\chi_{ij} \equiv \langle \sum_\sigma f_{i\sigma}^\dagger f_{j\sigma} \rangle$ and Lagrange multipliers λ_i to enforce the occupancy constraint on average, and Eq. (19) collapses to an effective free-fermion, or **spinon**, Hamiltonian,

$$\mathcal{H}_{\text{MF}} = - \sum_{ij} \frac{J_{ij} \chi_{ij}}{2} \sum_\sigma f_{i\sigma}^\dagger f_{j\sigma} + \text{h.c.} + \sum_i \lambda_i (n_i - 1). \quad (20)$$

This is precisely the same kind of self-consistency loop as Session 1’s Hubbard mean-field theory: guess χ_{ij} , diagonalize the resulting single-particle Hamiltonian, recompute χ_{ij} from the filled spinon orbitals, and repeat until convergence.

The payoff is that Eq. (20) is just a tight-binding model, and we already know how to read its spectrum. For the uniform chain, the self-consistent $\chi_{ij} \equiv \chi$ is the same on every bond, giving a simple cosine spinon band $\varepsilon(k) = -J\chi \cos k$, gapless Fermi points, and hence gapless spin excitations, matching everything we found directly from exact diagonalization in Section 3. For the dimerized chain of Eq. (14), the bond amplitude inherits the alternation of J_i , giving a dimerized, Su-Schrieffer-Heeger-like tight-binding chain for the spinons, whose spectrum opens a gap set by the difference between the strong- and weak-bond hopping amplitudes, mirroring the many-body gap of Section 7. The same construction generalizes to two dimensions, where an analogous mean-field ansatz on, say, the honeycomb lattice produces gapless **Dirac spinons**, the auxiliary-fermion language behind many candidate quantum spin liquids. None of this spinon construction is covered in the companion notebook; we flag it here as slides-and-lecture-only material, in the same spirit as the anisotropic exchange terms we flagged at the end of Session 1.

Summary

- Exact diagonalization means writing the Heisenberg Hamiltonian as a matrix on the full 2^L product-state basis and diagonalizing it directly, with no product-state restriction anywhere; the dimer's exact singlet ($-\frac{3}{4}J$) and triplet ($\frac{1}{4}J$) energies are the smallest worked example of this.
- The many-body gap $\Delta(L) = E_1 - E_0$ shrinks with system size; a power-law decay signals a gapless chain (the uniform Heisenberg chain, $\Delta \sim \pi^2 J/2L$), while saturation to a finite value, with exponentially fast finite-size convergence, signals a gapped chain (the dimerized chain).
- The XXZ model interpolates continuously between the classical, symmetry-broken Ising limit and the quantum, entangled Heisenberg limit; the finite-size splitting of the two lowest states already reveals the location of the exact, thermodynamic-limit phase transition.
- Total spin $\mathbf{S}_{\text{tot}}^2$ and S_{tot}^z are exactly conserved quantities of the Heisenberg Hamiltonian, for any L ; they block-diagonalize the exponentially large Hilbert space into much smaller sectors, they guarantee that a uniform field only splits levels linearly rather than mixing them, and they are exactly what pushes exact diagonalization out to $L \lesssim 20$ spins.
- A field-driven sequence of level crossings between successive total-spin sectors produces the magnetization staircase of a finite chain; impurity responses and non-local static correlators, decaying algebraically when gapless and exponentially when gapped, are single-snapshot diagnostics for exactly the same physics as the gap scaling.
- Dynamical correlators resolve excitations in both space and energy via the Lehmann representation; comparing spin- $\frac{1}{2}$ and spin-1 chains gives our first hint of the fractionalized, topological physics of the Haldane phase, developed properly with tensor networks in Session 4.
- Representing spins as constrained Abrikosov fermions and mean-field decoupling the exchange term gives an effective free-fermion, spinon, description of the excitations: gapless spinons for the uniform chain, gapped spinons for the dimerized chain, using exactly the same self-consistency machinery as Session 1's Hubbard mean-field theory.
- All of this is numerically exact, but the underlying cost is still exponential in L : even with every symmetry we can exploit, exact diagonalization caps out around a few tens of spins, which is precisely the wall that motivates the matrix-product-state methods of Session 4.